2018 IEEE SmartWorld, Ubiquitous Intelligence & Computing, Advanced & Trusted Computing, Scalable Computing & Communications, Cloud & Big Data Computing, Internet of People and Smart City Innovations

Design of remote database service system for wearable devices

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***Abstract*—Most wearable devices on the market today use Bluetooth short-distance transmission technology, and the working environment is inseparable from the smart phone, which has a lot of inconvenience. This paper is based on the existing wearable fall detection and positioning equipment, which embeds the SIM868 module and supports the function of the GPRS (General Packet Radio Service) remote access network. Based on wearable devices, this paper designs a remote database server system that integrates C/S (Client / Server) architecture and MySQL database. The system uses a TCP/IP (Transmission Control Protocol/Internet Protocol) protocol to bring together wearable devices, Android smart phones and the remote server. Both wearable devices and Android smart phones use a short connection to communicate with the server to increase the server's carrying capacity. The experimental test process is that the wearable device uploads the positioning information, and the information is stored in the database after being processed by the server. After the Android application accesses the server, the positioning information is displayed on the map to implement the positioning function. Through system evaluation, the server has a high user carrying capacity. Compared with B/S (Browser/Server) mode server, the remote database server based on C/S architecture and TCP/IP protocol has the advantages of low cost, reliable data transmission and fast response. With the gradual promotion of 5G technology and eSIM (embedded-SIM) technology, in the field of wearable devices, the system has a wide range of applications in the future.**

Keywords—Wearable, C/S, MySQL, TCP/IP, Database server

I. INTRODUCTION

With the rapid development of Internet of Things, big data, artificial intelligence, and communication technologies, more and more people use wearable medical devices to realize real-time monitoring of various physiological indicators of the human body [1][2][3]. According to market research firm Gartner, the health and medical field is recognized as the most promising area for wearable devices. In some health applications for wearable devices, monitoring data requires timely delivery of data packets, processing analysis and status feedback. For example, in the wearable monitoring application scenario for older people, offspring often need to remotely monitor their parents because they can't stay with the elderly at all times. Data remote transmission usually requires the participation of a database server. Moreover, the access of a large number of users is also a requirement of the server carrying capacity.

The most common wearable devices on the market today are implemented using Bluetooth technology. Wearable devices are used in combination with smart phones. The smart-phone is a medium for realizing access to the cloud server resources. The capabilities that wearable devices can achieve are very limited when they are away from the smart-phone. Therefore, it is more convenient to design a wearable device with independent communication capabilities. There are also smart watches with independent phone functions on the market, which is equivalent to a reduced version of the smart-phone. However, its functions are numerous and costly, and it is not suitable for the elderly to wear. The wearable device can realize remote communication by using the SIM module, saving money by using only data traffic. Considering the small amount of monitoring data that needs to be transmitted, GPRS technology can be used to access the remote database server. The GPRS technology is a packet bearing service developed on the basis of the GSM system. It currently provides stable data transmission of 20~40kb/s and supports real-time online [4].

This article designs a remote database server system, which is based on the existing wearable fall detection and positioning equipment. And its block diagram is shown in Fig. 1. The wearable device is embedded with a SIM module and supports the GPRS to access the Internet. Because currently in the network layer, almost all devices support the IP protocol. Therefore, in order to realize the function of data interaction between the wearable device client and the smart-phone client through the same server, this paper designs the server by TCP/IP protocol.

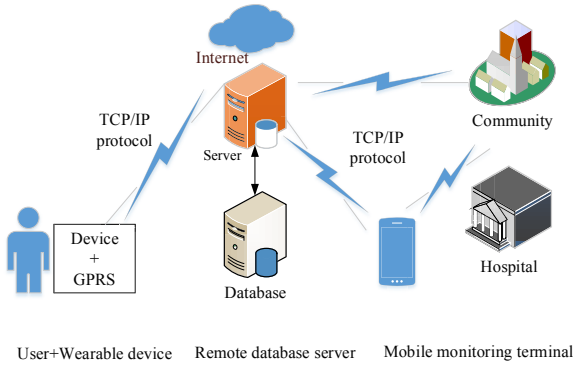

Figure1. the block diagram of the remote database server system

978-1-5386-9380-3/18/$31.00 ©2018 IEEE
DOI 10.1109/SmartWorld.2018.00060

II. METHOD

As can be seen from Fig. 1, the remote database server is the bridge from which data is transferred from the wearable device to the remote monitoring terminal. The wireless transmission technology uses the TCP/IP protocol. The server needs to have a one-to-many load capacity. In order to design a lightweight and reliable remote database server system, it is necessary to understand the architecture of software system, TCP/IP protocol and database server.

A. The Architecture of Software System

The remote monitoring system as shown in Fig. 1 is mainly divided into two modes in terms of software structure. (a) C/S mode (b) B/S mode.

1) C/S mode: C/S mode has a full-featured application on the client side that is responsible for interacting with the user and processing business logic. The C/S is generally established in a small-scale network environment. The LAN (local area networks) provides connection and data exchange services through a dedicated server. Commonly used configuration software in the monitoring system is usually C/S mode [5]. The C/S program can pay more attention to the process, check multiple levels of authority, and give less consideration to the running speed of the system. However, due to the integrity of the C/S program, it is necessary to investigate and deal with problems as a whole. The system upgrade is cumbersome and it may be a completely new system [5][6].

2) B/S mode: The B/S model mainly consists of a browser, a Web server, and a database server. The client adopts a browser. The interface is unified and the operation is simple, simplifying the functions of the client [5]. B/S is established on a wide area network and is generally accessible as long as there are operating systems and browsers. B/S has relatively weak control over security because it is built on a wide area network and is aimed at an unknown user base [5][6].

This paper designs a remote database server system based on C/S mode. C/S mode pays attention to process, the transmission path of data can be tracked in real time, and the storage and management function of data is more transparent.

B. The Protocol of TCP/IP

The commonly used network communication protocol in C/S mode is the TCP/IP protocol. Only TCP/IP allows full connection with the Internet. At present, the TCP/IP protocol is integrated in all operating systems [7]. TCP/IP is the basis of Internet interconnection, with scalability and reliability. In the OSI (Open System Interconnect) reference model, it abstracts the communication protocol into seven layers (because it is more complex and rarely used in practical applications) [7]. The simplified four-layer TCP/IP model includes the application layer, transport layer, Internet layer and network interface layer.

The transport layer TCP is a connection-oriented communication protocol. It establishes a connection through a three-way handshake and disconnects when the communication is completed. TCP provides a reliable data flow service and uses the "reaffirmation with retransmission" technology to achieve the reliability of the transmission. TCP packets include serial numbers and acknowledgments, so packages that are not received in sequence can be sorted, while damaged packages can be retransmitted [8]. UDP is a connectionless communication protocol. The UDP data includes destination port number and source port number information. Since communication does not require a connection, broadcast transmission can be achieved. UDP communication does not need to be confirmed by the receiver, which is an unreliable transmission, so packet loss may occur. UDP and TCP are on the same layer, but UDP does not care about the order, errors, or retransmission of packets. Therefore, UDP cannot be applied to connection-oriented services. Spoofing UDP packets is easier than spoofing TCP packets, so UDP-related services face greater risks.

The TCP/IP protocol used in this paper consists of the IP protocol of the network layer and the TCP protocol of the transport layer. TCP is more reliable than UDP, its three-way handshake mechanism is not easily misjudged by the receiver. The process of a client accessing a server based on the TCP protocol is shown in Fig. 2. First, the client establishes a short connection and sends string data. After the client receives the data returned by the server, it executes related commands or updates the relevant data, then closes the socket connection and the server releases the thread.

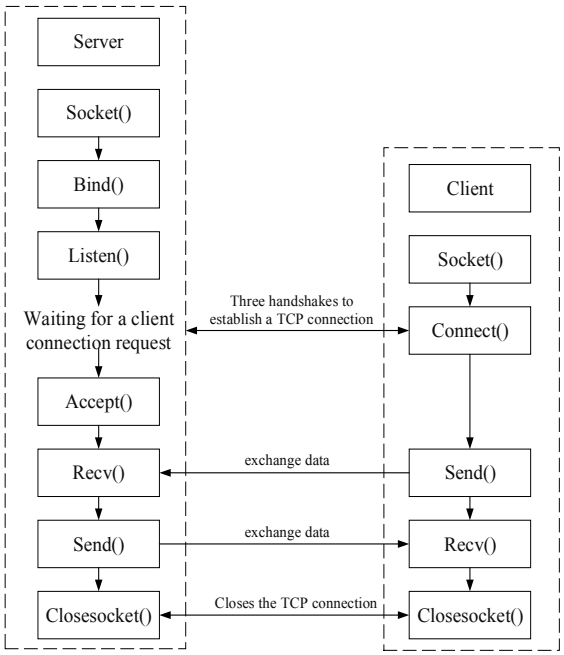

Figure2. TCP Communication Protocol in C/S Mode

C. Database Server

The database server consists of a MySQL database and a TCP server program. The server manages the database and completes functions such as data processing, addition, deletion, and modification.

1) MySQL database: It is one of the most popular relational database management systems. Relational databases

store data in different tables instead of all data in one large repository, which increases speed and flexibility. MySQL software, due to its small size, fast speed and low overall cost of ownership, especially the feature of open source, is generally chosen as the website database for the development of small and medium-sized websites [9].

Based on the existing wearable fall detection and positioning equipment, a total of three data tables were established in the initial database. The database needs to hold the binding relationship between the wearable device and the smart-phone, the monitoring data of the wearable device and the parameters set by the smart-phone. As shown in Fig. 3, the table of the Cellphone-Login is used to store the Bracelet-Numb, the Cellphone-Numb and Password, which contains all the information of the smart-phone registration and login as well as the binding relationship between the smart-phone and wearable device. In order to meet the requirement that a smart-phone can monitor multiple wearable devices at the same time, only the Bracelet-Numb is set as the primary key, which cannot be repeated when stored in the table. The primary key is not set in the Device-Update table because the wearable device periodically uploads the monitoring data. There may be multiple duplicate data, so the query is primarily based on the update time. In addition, the table of the Cellphone-Msg is similar to the table of the Device-Update for storing protocol data. When the smart-phone user logs in, the binding relationship is found through the table of the Cellphone-Login , and the required data is extracted from the table of the Device-Update.

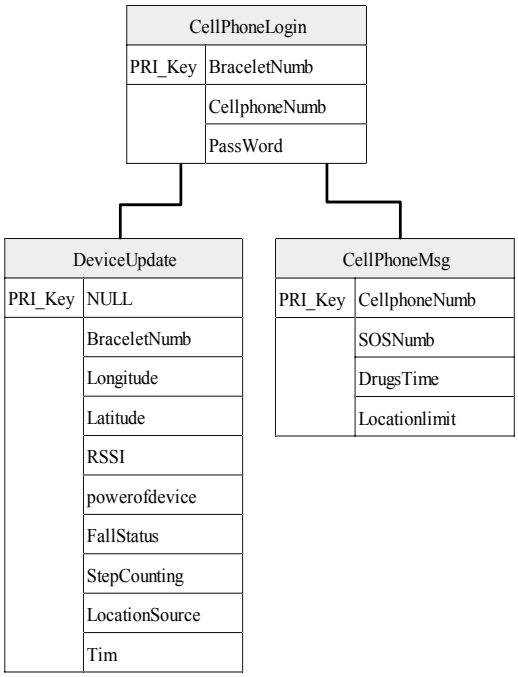

Figure3. tables build in the Database

2) TCP server architecture: The database runs on the same computer as the TCP server program, which includes all operations on the database. Due to the large number of clients and the need to be able to communicate with the server at any time, the server is required to have certain load capacity. The simplified client server communication architecture is shown in Fig. 4 below.

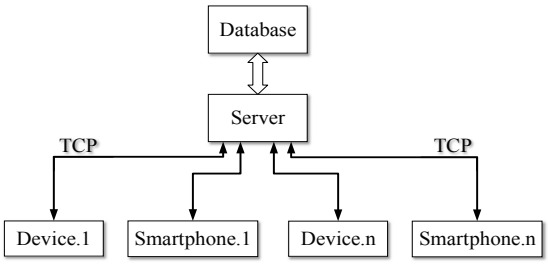

Figure4. the architecture of server communication system

As shown in Fig. 4, the client and the server are connected with the multiple-to-one method, and the server receives multiple threads. In theory, a 4G host can handle less than 4,000 threads. In order to solve the pressure of the server, the client uses the TCP short connection method, that is, the client establishes a TCP connection when sending data. Only when the data transfer is completed, the connection is released immediately, so that the client does not occupy the communication link for a long time. With a short connection, the server will theoretically have more than 4,000 users.

3) The communication protocol:

'protocol header' + 'id' + ',' + 'data'

The remote communication system includes the interaction of the wearable device client with the database server and the interaction of the Android client with the database server. For messages sent from different clients, the server handles different reception according to the data protocol header. In order to save the power consumption of the device sending, the wearable device is required to put all monitoring data into an array of strings each time it sends the data, and then send them all. The protocol header is parsed into a 2-digit decimal number so that the server can recognize the corresponding instruction. The communication protocol of the wearable device has only one data format, so the algorithm is not complicated. There are more communication protocols between the smart-phone and the server, including registration, login, query and other protocol data formats.

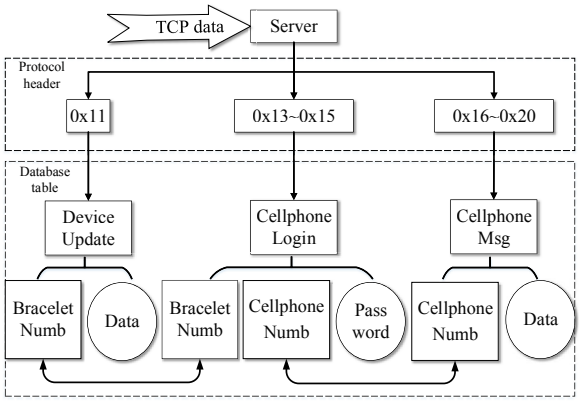

Figure5. the reception protocol of database server

As can be seen from Fig. 5, after the server receives the data, it first determines whether the protocol header is valid,

and then enters a different receiving thread for data processing. The table of the Cellphone-Login is used most frequently. If the Android client wants to obtain the data of the Device-Update table, it first completes the verification in the Cellphone-Login table and then queries the binding relationship. After obtaining the Bracelet-Numb, the required data will be queried from the Device-Update table.

III. Implementation

To achieve remote communication, the server needs to have an external network IP address. In this paper, Alibaba cloud server ECS is rented to provide the IP address of the external network. Both the server and database are running on the same host. The configuration of the host is shown in Table 1 below.

TABLE I. Server Host Configuration

CPU	*Memory*	*Band width*	*OS*	*System disk*
Intel Xeon E5-2682 v4	1G	1M	Windows Server 2012	40G

Implementation steps: Install the MySQL database on the host and configure the Visual Studio connection environment, because the server program is compiled using the Visual Studio 2015 C++ language. Configure the port number of the TCP server at Alibaba Cloud Console before debugging the program. The configuration file and the exe program are placed on the system disk and started to run.

Fig. 6 below is the flow of the wearable fall detection and positioning equipment access server used in this experiment. The hardware of the wearable device embeds a SIM868 module that can connect to the network using GPRS. The software program of the wearable device is embedded with an AT command based on the TCP communication protocol. GPS positioning data is preferred when the device obtains user location information. The wearable device will obtain the location information from the base station when the GPS signal is very poor. When the wearable device is activated, it accesses the Internet through the base station and accesses the database server based on the IP address. Finally, the client connects to the server's application via the port number. Monitoring data is periodically sent to the server when the wearable device is working properly.

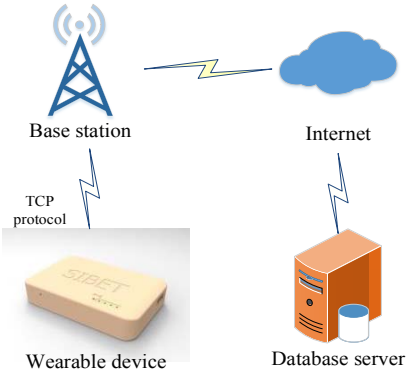

Figure6. the process of the wearable device accessing the server

IV. Result and Discussion

The remote database server system of this article is based on the existing wearable fall detection and positioning equipment. The 11-digit number of the SIM card was used as the ID during the experiment. After the wearable device starts up, it sends the positioning information to the server. The server accesses the base station database via the HTTP protocol, converts the received base station data into GPS data, and finally stores it in the database. The Android client access the server remotely over the network. Its application obtains the location information from the Device-Update table in the database and displays it on the Android interface.

Due to the use of C/S mode, the data transmission process can be tracked in real time. Check the relevant information from MySQL as shown in the following three figures.

BraceletNumb	CellphoneNumb	PassWord
188[redacted]16	182[redacted]60	12345679

Figure7. the table of the Cellphone-Login

CellphoneNumb	SosNumb	DrugsTime	Locationlimit
182[redacted]60	182[redacted]60	8:30-12:30-19:00	120.[redacted]21-121.[redacted]00-30.[redacted]30-32.[redacted]12

Figure8. the table of the Cellphone-Msg

Longitude	Latitude	BraceletNumb	RSSI	power	Statu	Step	Calorie	Sourse	Tim
120.4[redacted]	31.3[redacted]	188[redacted]16	C17	B1	D1	E0	F0	G1	2018/6/9 19:49
120.4[redacted]	31.3[redacted]	188[redacted]16	C18	B1	D1	E0	F0	G1	2018/6/9 19:50
120.4[redacted]	31.3[redacted]	188[redacted]16	C17	B1	D1	E0	F0	G1	2018/6/9 19:51
120.4[redacted]	31.3[redacted]	188[redacted]16	C13	B1	D1	E0	F0	G1	2018/6/9 20:25
120.4[redacted]	31.3[redacted]	188[redacted]16	C12	B1	D0	E0	F0	G1	2018/6/9 20:32
120.4[redacted]	31.3[redacted]	188[redacted]16	C13	B4	D0	E0	F0	G1	2018/6/9 23:16
120.4[redacted]	31.3[redacted]	188[redacted]16	C18	B4	D1	E0	F0	G1	2018/6/11 12:27
120.4[redacted]	31.3[redacted]	188[redacted]16	C18	B4	D1	E0	F0	G1	2018/6/11 12:27
120.4[redacted]	31.3[redacted]	188[redacted]16	C18	B3	D1	E0	F0	G1	2018/6/11 12:48
120.4[redacted]	31.3[redacted]	188[redacted]16	C16	B2	D1	E0	F0	G1	2018/6/11 13:59
120.4[redacted]	31.3[redacted]	188[redacted]16	C16	B2	D1	E0	F0	G0	2018/6/11 14:00

Figure9. the table of the Device-Update

A lightweight Android mobile application was developed for remote testing based on a custom data transfer protocol. Combined with the Gaode map API, Fig. 10 shows the location information obtained from the database server.

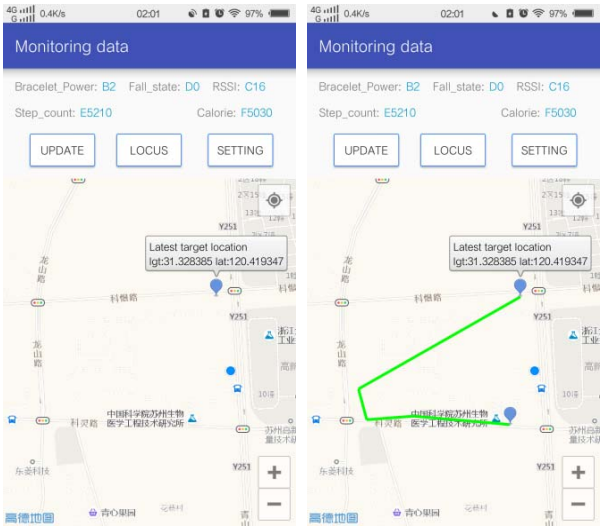

Figure10. the positioning function of the android app

As shown in Fig. 10, the android app displays the location and trajectory of the target in the map. The communication protocol on the smart-phone corresponds to the server reception protocol shown in Fig. 5. After the mobile terminal accesses the server, the server finds the latest location information from the data table shown in Fig. 9 and displays the result in the map.

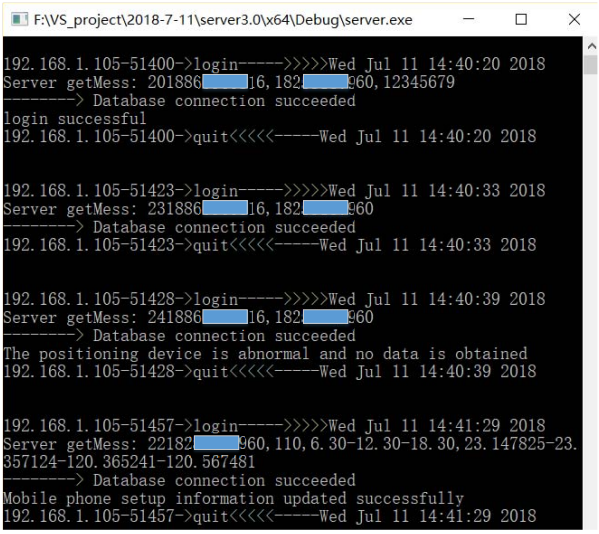

Figure11. the server monitoring interface

The server program is a win32 program that can monitor the working status of the server in real time, And the overall UI (user interface) is a black form. Fig. 11 shows the interface of the server monitor.

As mentioned above, the communication protocol designed in this paper is based on short-time connection. As can be seen from the client access record of Fig. 11, the client completes the process from login to logout within 1 second. The system is evaluated according to the carrying capacity of the server, and the number of clients that the server can receive is related to the configuration of the server resources. The configuration of the server leased in this article is shown in Table 1. A 1G memory host can theoretically receive 1024 customers in one second, but considering the memory usage of Windows systems and MySQL, the evaluation result is up to 600 customers per second. It can be calculated from the data table in Fig. 3 that the amount of data transmitted each time is small, and the maximum amount of data is less than 100B. Calculated by the maximum number of users, the data transfer rate needs at least 60KB/s. In the case that the 1M bandwidth can meet the data transmission rate of 128KB/s, the system can support 600 customers to send and receive data at the same time. As can be seen from the literature [10], the earlier version of MySQL supports 20,000 queries per second and more than 10,000 connection threads, when the amount of data is very small. For this experiment, the database can easily satisfy 600 user threads at the same time. In addition, there is no peak period in the system, and the wearable device uploads data dynamically and periodically. The data upload period is shortened to 30 seconds only when the user has a fall behavior. Under the assumption that the client accesses the server once in 30 seconds, the user can expand at least 30 times to reach 18,000 users.

The system adopts C/S architecture and TCP communication protocol, and has the following advantages compared with servers adopting B/S architecture and HTTP protocol [11] [12]. (a) The B/S structure is a typical 3-layer structure. Although the user's operation is simple, it is too burdensome for the server. In order to implement the same application, the B/S mode server takes up more resources. (b) The HTTP protocol of the application layer is not as reliable as the TCP protocol of the transport layer. For administrators, using a browser to perform system maintenance is very insecure and inconvenient [13]. (c) The database server system designed in this paper is based on lightweight data transmission, and its data transmission and response speed is fast. Based on the application layer's HTTP protocol, data transmission and reception need to select a specific format to parse, and the response speed is slow.

V. Conclusions

In the past, most wearable devices were combined with Bluetooth technology and mobile phones, requiring that the device's Bluetooth was always working, which would cause great trouble. With the advent of eSIM [14] and 5G, wearable devices with independent communication capabilities are more competitive in the market. In this paper, the remote database server of C/S mode combined with TCP/IP protocol is designed to satisfy all communication modes that support IP protocol. It can be seen from the database tracking that the system communication quality is good and the data protocol is transparent. The system's carrying capacity has been evaluated in terms of memory and bandwidth. At present, the lowest configured server resources on the market can theoretically meet 18,000 customer access needs. If it is necessary to increase the number of users, you can rent a higher configured server resource or use a server cluster. In addition, compared with servers based on B/S architecture and HTTP protocol, the system has the advantages of low server cost, high reliability and fast response in remote data transmission and positioning applications based on wearable devices.

Acknowledgment

We gratefully acknowledge the support from the ministry of science and technology international cooperation project(2016YFE0107700).

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